

Two Variables Are Not Enough

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Abstract. Let n be the smallest integer such that every closed lambda term beta converts to one with at most n bound variables. We show that $n = 3$.

We shall work in the untyped lambda calculus. For the most part we shall adopt the notation and terminology of [1].

It is well known that every closed lambda term beta converts to one with only three distinct bound variables. This can be seen by converting each closed lambda term into an applicative combination of S 's and K 's. It is equally clear that not every closed term can be beta converted into one with only one bound variable; since, such terms are closed under beta reduction (without the aid of alpha conversion). Here we settle the case of two variables in the negative.

As a consequence we conclude the following.

It is easy to see the mini-max principle holds (modulo insertions of K); the minimum number of bound variables needed to write a closed term equals the maximum number of free variables in a subterm. Thus any closed term beta convertible to S must contain a subterm with three distinct free variables. The same holds true for B , since B, C^*, K , and W^* form a basis ([1] pg 210).

We shall define a combinatory logic suitable for discussing terms with at most two bound variables. To fix ideas we shall fix these variables to be x, y . The combinatory logic consists of atoms (constants, combinators)

A

together with reduction rules

$\rightarrow$.

We call the logic HOT (combinators Hereditarily of Order Two). The atoms and rules are defined by a simultaneous recursion as follows.

If X is an applicative combination of x 's and y 's then the HOT atom A is introduced with the reduction rule

$$Axy \rightarrow X$$

and the rank of A is 1.

If X is an applicative combination of x 's, y 's, and HOT atoms of ranks $< n$ then the HOT atom A is introduced with the reduction rule

$$Axy \rightarrow X$$

and the rank of A is n .

For convenience below we shall assume that HOT atoms of rank n also have rank $n + 1$ (as if the above recursive definitions were not given simultaneously).

Each HOT combination X of x 's, y 's, and atoms corresponds to a lambda term X^* by

$$x^* = x$$

$$y^* = y.$$

If A has reduction rule $Axy \mapsto X$ then $A^* = \lambda xy.X^*$

$$(XY)^* = (X^* Y^*).$$

It will be convenient to use the notation $\rightarrow$ for beta reduction and $\leftrightarrow$ for beta conversion of lambda terms.

Suppose that X is a lambda term using only the variables x, y (possibly both free and bound). Then there exists a HOT combination H such that $H^* \rightarrow X$. Define the lambda term $X+$ by

$$x+ = x$$

$$y+ = y$$

$$(XY)+ = (X + Y+)$$

$$\begin{aligned} (\lambda x.X)+ &= (\lambda yx.X+)y \text{ if } y \text{ is free in } X \\ &(\lambda yx.X+)K \text{ if } y \text{ is not free in } X \\ (\lambda y.X)+ &= (\lambda xy.X+)x \text{ if } x \text{ is free in } X \\ &(\lambda xy.X+)K \text{ if } x \text{ is not free in } X \end{aligned}$$

Then $X+ \rightarrow X$ and $X+$ alpha converts (by the use of a third variable) to $X^\wedge$ defined by

$$x^\wedge = x$$

$$y^\wedge = y$$

$$(XY)^\wedge = (X^\wedge Y^\wedge)$$

$$\begin{aligned} (\lambda x.X)^\wedge &= (\lambda xy.[x/y, y/x]X^\wedge)y \text{ if } x \text{ is free in } X \\ &(\lambda xy.[x/y, y/x]X^\wedge)K \text{ if } y \text{ is not free in } X \\ (\lambda y.X)^\wedge &= (\lambda xy.X^\wedge)x \text{ if } x \text{ is free in } X \\ &(\lambda xy.X^\wedge)K \text{ if } x \text{ is not free in } X. \end{aligned}$$

From $X^\wedge$ we can define H as $\#X^\wedge$ where

$$\#x = x$$

$$\#y = y$$

$$\#(XY) = (\#X\#Y)$$

$$\#(\lambda xy.X) = A \text{ where } A \text{ has the reduction rule } Axy \mapsto \#X.$$

HOT combinators together with $\multimap$ form a regular left normal combinatory reduction system ([2]) and therefore satisfy the Church-Rosser theorem and standardization theorem. $\multimap_1$ stands for leftmost- outermost reduction ([1] pg. 323). $\multimap_{\text{head}}$ stands for head reduction ([1] pg 169).

A closed HOT combination M is said to be “head secure” if there is a combination Y containing y such that $Mxy \multimap xY$. The combination X is said to “unroll” to Y if there are HOT combinators (atoms) $A_1, \dots, A_n$ (here we allow $n = 0$) such that $X \multimap_1 A_1(\dots(A_n Y) \dots)$. Such Y are clearly closed under head reduction.

Lemma: If M is a head secure combination of HOT combinators of rank $< n+1$ and $Mxy \multimap_{\text{head}} X$ then X is a combination of y ’s, HOT combinators of rank $< n$ and Y ’s such that Mx unrolls to Y .

Proof: By induction on the length of a head reduction to X . For the basis case Mx unrolls to Mx . Now suppose that $Mxy \multimap_{\text{head}} X1 \multimap_{\text{head}} X2$ and the conclusion is true of $X1$. We distinguish two cases.

Case 1: $X1 = YY1 \dots Ym$ where Mx unrolls to Y . Since Mxy is head secure Y is not an atom by Church-Rosser. If Y begins with a head redex then the conclusion holds for $X2$. Otherwise $Y = AZ$ and by definition Mx unrolls to Z . Moreover if A' appears on the r.h.s of the reduction rule for A then A' has rank $< n$. Thus in this case $X2$ satisfies the lemma.

Case 2: $X1 = AY1 \dots Ym$ where rank $A < n$. As above $X2$ satisfies the lemma.

Corollary: If M is head secure then Mx has a normal form $A1(\dots(Anx)\dots)$.

Proof: $Mxy \multimap_{\text{head}} xY$ where y belongs to Y . Since Mx does not unroll to xY it must unroll to x . Define Hn by

$$H0xy \multimap x(yy)$$

$$H(n+1)xy \multimap x(y(Hn)).$$

Proposition: There is no combination M of HOT combinators of rank $< n+1$ such that $M^* \longleftrightarrow Hn^*$.

Proof: By induction on n , the basis case when $n = 0$ is trivial by definition. Suppose that $n > 0$ and that $M^* \longleftrightarrow Hn^*$. Then $Mxy \multimap x(yH(n-1)^*)$ and by the standardization theorem there exists a reduction

$$M^*xy \multimap_{\text{head reduction}} xX$$

$$X \multimap_{\text{head reduction}} xY$$

$$Y \multimap_{\text{standard}} H(n-1)^*.$$

Thus we have

$$\begin{aligned} Mxy &\succ\text{head reduction}\rightarrow x\#X \\ \#X &\succ\text{head reduction}\rightarrow y\#Y. \end{aligned}$$

In particular, M is head secure so Mx has a normal form

$$A1(\dots(Akx)\dots).$$

We have then by Church-Rosser and standardization

$$\begin{aligned} (@) \quad A1(\dots(Akx)\dots)y &\succ\text{head}\rightarrow xZ \\ Z &\succ\text{head}\rightarrow yW. \end{aligned}$$

where $W^* \rightarrow H(n-1)^*$. Now W is a combination of HOT combinators of rank $< n$, y 's and terms $Ai(\dots(Akx)\dots)$ for $i = 1, \dots, k+1$. Put $N = [OMEGA/x, OMEGA/y]W$. We distinguish two cases.

Case 1: $n = 1$. Then $Nxy \rightarrow x(yy)$.

Case 2: $n > 1$. Then $Nxy \rightarrow xX1 \rightarrow x(yY1)$ where $Y1^* \rightarrow H(n-2)^*$. In either case N is head secure so Nx has a normal form

$$A'1(\dots(A'mx)\dots).$$

Now in case 1, each HOT combinator in N appears in function position iteratively applied to OMEGA. Thus we have that $m = 0$ and the reduction $Nxy \rightarrow x(yy)$ is impossible. In the second case, by similar reasoning, rank $(A'j) < n$ for $j = 1, \dots, m$. By Church-Rosser and standardization there exists a reduction

$$\begin{aligned} A'1(\dots(A'mx)\dots)y &\succ\text{head}\rightarrow xX2 \\ X2 &\succ\text{head}\rightarrow yY2 \end{aligned}$$

where $Y2^* \rightarrow H(n-2)^*$. This reproduces (@) at lower rank.

Corollary: There is no HOT combination M such that $M^* \longleftrightarrow S$ (or B).

References

- [1] Barendregt, The Lambda Calculus, North Holland 1984.
- [2] Klop, Combinatory Reduction Systems Math. Centrum Amsterdam 1980.
- [3] Statman, Combinators Hereditarily of Order Two CMU Math. Dept. Technical Report 88-33, August 1988.